1. What will be the output of the following Python code?

```
x = 5
y = x
x = 10
print (y)
```

Options:

A. 5
B. 10
C. 15
D. Error

Correct Answer: A. 5

2. Assume the user enters 4. What will be the output?

```
x = input()
y = int(x) + 2
print(type(x), y)
```

Options:

A. <class 'int'> 6
B. <class 'str'> 6
C. <class 'str'> 42
D. Error

Correct Answer: B. <class 'str'> 6

3. What will be the output?

```
print("Data","Science", sep="-", end=" ")
print("Test")
```

Options:

A. Data Science
B. Data-Science Test
C. Data-Science
D. Data Science Test

Correct Answer: B. Data-Science Test

4. What will be the output?

```
print(list(range(2,10,3)))
```

Options:

A. [2,5,8]
B. [2,4,6,8]
C. [3,6,9]
D. [2,5,8,11]

Correct Answer: A. [2,5,8]

5. What will be the output?

```
for i in range(5):
    if i == 2:
        continue
    if i == 4:
        break
    print(i, end=" ")
```

Options:

A. 0 1 2 3
B. 0 1 3
C. 0 1 2 3 4
D. 0 1 3 4

Correct Answer: B. 0 1 3

6. What will be the output?

```
func = lambda x, y: x + y * 2
print(func(3,4))
```

Options:

A. 14
B. 11
C. 10
D. Error

Correct Answer: B. 11

7. What will be the output?

```
a = 5
b = 2.0
c = a + b
print(type(c))
```

Options:

A. <class 'int'>
B. <class 'float'>
C. <class 'double'>
D. Error

Correct Answer: B. <class 'float'>

8. What will be the output?

```
x = "10"
y = 5
```

```
z = int(x) + y
print(z, type(z))
```

Options:

A. 105 <class 'str'>
B. 15 <class 'int'>
C. 15 <class 'float'>
D. Error

Correct Answer: B. 15 <class 'int'>

9. What will be the output?

```
x = "3.5"
y = int(float(x))

print(y, type(y))
```

Options:

A. 3 <class 'int'>
B. 3.5 <class 'float'>
C. 3 <class 'float'>
D. Error

Correct Answer: A. 3 <class 'int'>

10. Which Python data type is immutable?

Options:

A. List
B. Dictionary
C. Tuple
D. Set

Correct Answer: C. Tuple

11. What will be the output?

```
text = "data science"
result = text.title().replace("Science", "Analysis")
print(result)
```

Options:

A. Data science
B. Data Science
C. Data Analysis
D. data analysis

Correct Answer: C. Data Analysis

12. What will be the output?

```
s = "Python"
result = s[1] + s[-1] + s.upper()[0]
print(result)
```

Options:

A. yNP
B. ynP
C. ynp
D. yNPython

Correct Answer: A. yNP

13. What will be the output?

```
s = "Data"
print(s[-1] + s[-3])
```

Options:

A. at
B. ta
C. aa
D. Error

Correct Answer: A. at

14. What will be the output?

```
my_list = [1,2,3]
my_list.append([4,5])
print(len(my_list))
```

Options:

A. 3
B. 4

C. 5
D. Error

Correct Answer: B. 4

15. What will be the output?

```
a = [1,2,3]
b = a
b.append(4)
print(a)
```

Options:
A. [1,2,3]
B. [1,2,3,4]
C. [4,1,2,3]
D. Error

Correct Answer: B. [1,2,3,4]

16. What will be the output?

```
nums = [1,2,3,4]
result = [x*2 for x in nums if x%2==0]
print(result)
```

Options:
A. [2,4,6,8]
B. [4,8]
C. [2,6]
D. [1,4,9,16]

Correct Answer: B. [4,8]

17. What will be the output?

```
a = [[0]*3]*3
a[0][1] = 5
print(a)
```

Options:
A. [[0,5,0],[0,0,0],[0,0,0]]
B. [[0,5,0],[0,5,0],[0,5,0]]
C. [[0,0,0],[0,5,0],[0,0,0]]
D. Error

Correct Answer: B. [[0,5,0],[0,5,0],[0,5,0]]

18. What will happen when this code runs?

```
t = (1,2,3)
t[0] = 10
print(t)
```

Options:

A. (10,2,3)
B. (1,2,3)
C. TypeError
D. (10,1,2,3)

Correct Answer: C. TypeError

19. What will be the output?

```
d = {'a':1,'b':2}
d['c'] = d.get('a',0) + d.get('d',3)
print(d)
```

Options:

A. {'a':1,'b':2,'c':1}
B. {'a':1,'b':2,'c':4}
C. {'a':1,'b':2,'c':3}
D. Error

Correct Answer: B. {'a':1,'b':2,'c':4}

20. What will be the output?

```
data = {'a':10,'b':20,'c':30}
value = data.pop('b')
print(value, data)
```

Options:

A. 20 {'a':10,'b':20,'c':30}
B. 20 {'a':10,'c':30}
C. {'a':10,'c':30}
D. Error

Correct Answer: B. 20 {'a':10,'c':30}

21. What will be the output?

```
data = {'x':[1,2],'y':[3,4]}
data['x'].append(5)
print(len(data['x']) + len(data['y']))
```

Options:

A. 3
B. 4
C. 5
D. 6

Correct Answer: C. 5

22. What will be the output?

```
s = {1,2,2,3,4,4}
print(len(s), sorted(s))
```

Options:

A. 6 [1,2,2,3,4,4]
B. 4 [1,2,3,4]
C. 4 {1,2,3,4}
D. Error

Correct Answer: B. 4 [1,2,3,4]

23. Which Python data structure does not allow duplicate elements?

Options:

A. Set
B. String
C. Dictionary
D. Tuple

Correct Answer: A. Set

24. What will be the output?

```
import numpy as np
a = np.array([1,2,3])
print(type(a))
```

Options:

A. <class 'list'>
B. <class 'numpy.ndarray'>
C. <class 'array'>
D. Error

Correct Answer: B. <class 'numpy.ndarray'>

25. What will be the output?

```
import numpy as np
a = np.array([[1,2,3],[4,5,6]])
print(a.shape)
```

Options:
A. (2,3)
B. (3,2)
C. (6,)
D. (2,)

Correct Answer: A. (2,3)

26. What will be the output?

```
import numpy as np
a = np.array([10,20,30,40])
print(a[2])
```

Options:
A. 20
B. 30
C. 40
D. Error

Correct Answer: B. 30

27. What will be the output?

```
import numpy as np
a = np.array([1,2,3,4,5])
print(a[1:4])
```

Options:
A. [2 3 4]
B. [1 2 3 4]
C. [2 3 4 5]
D. [1 2 3]

Correct Answer: A. [2 3 4]

28. What will be the output?

```
import numpy as np
a = np.array([1,2,3])
print(a*2)
```

Options:

A. [2 4 6]
B. [1 4 9]
C. [1 2 3 1 2 3]
D. Error

Correct Answer: A. [2 4 6]

29. What will be the output?

```
import numpy as np
a = np.arange(1,10,3)
print(a)
```

Options:

A. [1 4 7]
B. [1 3 6 9]
C. [1 2 3 4 5 6 7 8 9]
D. [1 4 7 10]

Correct Answer: A. [1 4 7]

30. What will be the output of the following Python code?

```
import numpy as np
a = np.arange(3)
b = np.ones(3)
c = np.zeros(3)
d = np.full(3,5)

print(a + b + c + d)
```

Options:

A. [6. 7. 8.]
B. [5. 6. 7.]
C. [6 7 8]
D. Error

Correct Answer: A. [6. 7. 8.]